

Curriculum Vitae Caluã de Lacerda Pataca

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Former Fulbright fellow and Ph.D. graduate from the Rochester Institute of Technology (RIT) with research at the intersection of Human-Computer Interaction (HCI) and Computing Accessibility.

EDUCATION

Rochester Institute of Technology (RIT), Rochester, New York, United States

Ph.D. in Computing and Information Sciences

08/2021–10/2025

Advisors: Dr. Matt Huenerfauth & Dr. Roshan Peiris

Thesis: Improving the Accessibility of Speech for Deaf and Hard-of-Hearing Individuals Through Affective Captions

University of Campinas (Unicamp), Campinas, São Paulo, Brazil

M.Sc. in Electrical and Computer Engineering

08/2018–05/2021

Advisor: Dr. Paula Dornhofer Paro Costa

Thesis: Speech-Modulated Typography

B.A. and Licentiate degree in Visual Arts

02/2002–07/2007

Certificate program in programming (technical high school from Unicamp)

02/1999–12/2002

PEER-REVIEWED CONFERENCE PAPERS

- [C9] **de Lacerda Pataca, Caluã***, SooYeon Ahn*, Suhyeon Yoo*, JooYeong Kim, Khai Truong, Jin-Hyuk Hong, Roshan Peiris, and Matt Huenerfauth. ["CuCap: Comparative Analysis of Customized Captioning for English and Korean d/Deaf and Hard-of-Hearing Users."](#) In *Proceedings of the 27th International ACM SIGACCESS Conference on Computers and Accessibility*, pp 1–20. 2025.
- [C8] Jeremy Zhengqi, **Caluã de Lacerda Pataca**, Liang-Yuan Wu, and Dhruv Jain. ["CapTune: Adapting Non-Speech Captions With Anchored Generative Models"](#) In *Proceedings of the 27th International ACM SIGACCESS Conference on Computers and Accessibility*, pp 1–18. 2025.
- [C7] **de Lacerda Pataca, Caluã**, Saad Hassan, Lloyd May, Michelle Olson, Toni D'Aurio, Roshan Peiris, and Matt Huenerfauth. ["Tactile Emotions: Multimodal Affective Captioning with Haptics Improves Narrative Engagement for d/Deaf and Hard-of-Hearing Viewers."](#) In *Proceedings of the 2025 ACM CHI Conference on Human Factors in Computing Systems*, pp. 1–17. 2025.
- [C6]  **de Lacerda Pataca, Caluã**, Saad Hassan, Nathan Tinker, Roshan Peiris, and Matt Huenerfauth. ["Caption Royale: Exploring the Design Space of Affective Captions from the Perspective of Deaf and Hard-of-Hearing Individuals."](#) In *Proceedings of the 2024 ACM CHI Conference on Human Factors in Computing Systems*, pp. 1–17. 2024. Honorable mention.

* Denotes equal contribution.

- [C5]  Hassan, Saad, **Caluã de Lacerda Pataca**, Laleh Nourian, Garreth Tigwell, Briana Davis, and Will Wagon. ["Designing and Evaluating an Advanced Dance Video Comprehension Tool with In-situ Move Identification Capabilities."](#) In *Proceedings of the 2024 ACM CHI Conference on Human Factors in Computing Systems*, pp. 1–19. 2024. Honorable mention.
- [C4] **de Lacerda Pataca, Caluã**, Matthew Watkins, Roshan Peiris, Sooyeon Lee, and Matt Huenerfauth. ["Visualization of Speech Prosody and Emotion in Captions: Accessibility for Deaf and Hard-of-Hearing Users."](#) In *Proceedings of the 2023 ACM CHI Conference on Human Factors in Computing Systems*, pp. 1–15. 2023.
- [C3]  Hassan, Saad*, Akhter Al Amin*, **Caluã de Lacerda Pataca***, Diego Navarro, Alexis Gordon, Sooyeon Lee, and Matt Huenerfauth. ["Support in the Moment: Benefits and Use of Video-Span Selection and Search for Sign Language Video Comprehension among ASL Learners."](#) In *Proceedings of the 24th International ACM SIGACCESS Conference on Computers and Accessibility*, pp. 1–14. 2022. Best paper nomination.
- [C2] **de Lacerda Pataca, Caluã**, and Paula Dornhofer Paro Costa ["Speech modulated typography: towards an affective representation model."](#) In *Proceedings of the 25th International Conference on Intelligent User Interfaces*, pp. 139–143. 2020.
- [C1] **Caluã de Lacerda Pataca**, and Paula Dornhofer Paro Costa ["Tipografia modulada pela fala: avaliação de um algoritmo de geração de prosódia visual em textos."](#) In *Blucher Design Proceedings* 6, no. 4. 2019.

JOURNAL ARTICLES

- [J2] Hassan, Saad, **Caluã de Lacerda Pataca***, Laleh Nourian*, Akhter Al Amin*, Diego Navarro, Sooyeon Lee, Alexis Gordon, Matthew Watkins, Garreth Tigwell, and Matt Huenerfauth. ["Exploring the Benefits and Applications of Video-Span Selection and Search for Real-Time Support in Sign Language Video Comprehension among ASL Learners."](#) *ACM Transactions on Accessible Computing* 17, no. 3 (2024): 1–35.
- [J1] **de Lacerda Pataca, Caluã**, and Paula Dornhofer Paro Costa. ["Hidden Bawls, Whispers, and Yelps: Can Text Convey the Sound of Speech, Beyond Words?"](#) *IEEE Transactions on Affective Computing* 14, no. 1 (2022): 6–16.

PATENTS

- Paula Dornhofer Paro Costa, and **Caluã de Lacerda Pataca**. Método para composição de closed-captions com parâmetros tipográficos representando visualmente qualidades acústicas da fala (Pending. Process BR-10/2021-016125-6).
- Ricardo Alexandre Da Silva Shimabukuro Victorio, Gustavo Vilela Vargas, Fernando Augusto Chufi, Thiago Yoshihiro Ito Takahashi, Fábio San Juan, **Caluã de Lacerda Pataca**, José Luis Ponciano Bomfim, Adriana Ramos, Guilherme de Oliveira Feliciano, and Maria Cristina Marcucci Ribeiro (2012). Traceability system applied to farm production activities, industrialization and commercialization of bee products. U.S. Patent Application No. 12/596,612.

INVITED TALKS & GUEST LECTURES

- 09/25 Adding Nuance: Emotion-Aware, Flexible Captions for Speech Access · State of the Science Conference at Gallaudet University
- 02/25 Improving Access, Satisfaction, and Engagement with Video through Affective Captions · William & Mary Department of Computer Science
- 09/24 Captions beyond words: Paralinguistic accessibility for Deaf and Hard-of-Hearing individuals · 'Artificial Intelligence for Multimodal Signal Processing' Lab at the University of Campinas School of Electrical and Computer Engineering
- 03/24 Captions beyond words: Paralinguistic accessibility for Deaf and Hard-of-Hearing individuals · Department of Access Services, RIT
- 11/23 Enhancing Affective Captions with Haptic Feedback · Haptics for Inclusion Symposium, University of Borås, Sweden
- 10/23 Captions beyond words: Paralinguistic accessibility for d/Deaf and Hard-of-Hearing individuals · Communication Sciences and Disorders Department, University of Houston
- 03/23 Visualization of Speech Prosody and Emotion in Captions: Accessibility for Deaf and Hard-of-Hearing Users · ISTE-266 Design for Accessibility, RIT
- 09/22 Identifying emotions, moods, and emphasis through captions · Readability Consortium
- 06/13 Design Issues in Web Application and Website Projects · Semana de Atualização em Informática, UTFPR (Ponta Grossa), Brazil, presented together with Lucas Vignoli Reis

TEACHING

- **Teaching Assistant for the ISTE-798 Future Interactions course** 2025
Dr. Garreth Tigwell's class at the Rochester Institute of Technology
- **Experimental Research Design Lecture** 2022 & 2024
HCIN-600 Research Methods class under Dr. Roshan Peiris at the Rochester Institute of Technology

MENTORSHIP & SUPERVISION

Supervised 3 Master's and 4 undergraduate students (2022–2025)

- *Stephanie Patterson* (MSc HCI, 2025–): Data collection for two of my studies; advising independent study; co-author on pending and planned submissions.
- *Sidney Grabosky* (MSc HCI, 2025–): Game design and study; co-author on upcoming paper submission.
- *Matthew Watkins; Michelle Olson* (MSc HCI, 2022; 2024): Study coordination and participant recruitment across multiple lab and remote sessions.

- *Yiyuan (Steve) Wufeng* (BS CS, 2023–24): Mentored through internship and co-op; advised student-led study using our jointly developed captioning pipeline.
- *Andrew Chacon* (BS CE, 2023): Supervised development of caption-rendering pipeline for my research program.
- *Russell Lee* (BS HCC, 2023): Supervised software projects including open-source jsPsych plugin (pending contribution).
- *Nathan Tinker; Toni D'Aurio* (BS ASL-English Interpretation, 2022; 2024): Coordinated ASL video recording for dictionary project; supervised recruitment and data collection.

SCHOLARSHIPS & AWARDS

- 2021–2026 Fulbright/Capes Full Doctorate Program · *The highly competitive Fulbright Scholarship program supports doctoral study at top U.S. institutions. (\$120,000)*
- Two honorable mentions at CHI'24 · *Top 5% of conference papers.*
- Best paper nomination at ASSETS'22 · *Top 6% of conference papers.*
- 2022 CRA-WP Grad Cohort Workshop for IDEALS · *This competitive program for students under-represented in the computing field provides mentoring for graduate students across North America.*
- SIGCHI Student Travel Grant (SSTG) to present at the 2020 International Conference on Intelligent User Interfaces at Cagliari, Italy · *Competitive program providing support to students attending SIGCHI-sponsored or co-sponsored conferences.*
- 3 bronze medals, 1 silver medal, and 2 Popular Vote Trophies in 2018, 2019, and 2020 editions of the Brasil Design Award · *The awards were presented at the annual ceremony hosted by the Association for Brazilian Design Companies (ABEDESIGN).*
- Selected project for the 13th Brazilian Graphic Design Biennial · *Prestigious juried exhibition organized by the Brazilian Association of Graphic Designers.*

PROFESSIONAL EXPERIENCE

Graduate Research Assistant Rochester Institute of Technology

08/2021–12/2025

- Research on DHH-centered captioning approaches: affective, prosodic, and haptic
- Mixed-methods research design: surveys, interviews, usability studies
- Prototype development: interactive systems and visualizations (e.g., [C1], [C3], [C4])
- Academic writing: papers for HCI and accessibility venues
- Student supervision: mentoring and coordinating RAs, interns, co-ops across ML, conferencing, and captioning projects

- Communications Chair (CAIR Lab, 2022–2024): coordinating with guests for presentations at the lab, along with social media outreach about

Founder and Head of Design
Preface (Brazil)
10/2006–08/2021

- Led UX, UI, front-end, and branding design across mid-large-scale projects
- Delivered national-impact platforms; awarded multiple national design prizes (e.g., Brazilian History Olympiad with 200k+ users annually)

PROFESSIONAL SERVICE

Associate Chair (AC) for the Late-Breaking Work Track
2025

ACM Conference on Human Factors in Computing Systems (CHI'25)

Web and Graphic Design Chair
2024–25

ACM SIGACCESS Conference on Computers and Accessibility (ASSETS'25)

Technical Paper Reviewer
2022–Present

ACM Conference on Human Factors in Computing Systems (CHI'23–26)

Technical Paper Reviewer
2025

IEEE Conference on Virtual Reality and 3D User Interfaces (IEEE VR 26)

Late Breaking Work Reviewer
2023

ACM Conference on Human Factors in Computing Systems (CHI'23)

Technical Paper Reviewer
2021

Congresso Nacional de Iniciação Científica em Design (CONGIC)

PROFESSIONAL DEVELOPMENT

- **Interacting with Computers: Reviewing Academy**, with Carine Lallemand & Helen Petrie2025
- **The Expressive Function of Prosody in Speech**, with Plínio A. Barbora2019
- **Scientific Writing Workshop**, with Inês Signorini2019
- **Introduction to Computational Design**, with Guilherme Vieira2017
- **Graphic Production**, with Aline Valli2013
- **Branding and Culture**, with André Stolarski2012

POSTER PRESENTATIONS

[P3]

Caluã de Lacerda Pataca, Matthew Watkins, Sooyeon Lee, and Matt Huenerfauth. (2022) Devel-
oping a captioning style that can help Deaf and Hard-of-hearing individuals identify a speaker's
emotions, moods, and emphasis. *AI @ RIT*

- [P2] **Caluã de Lacerda Pataca**, and Matt Huenerfauth. (2022) Visualizing paralinguistic dimensions of speech through typography. *2022 CRA - WP Grad Cohort Workshop for IDEALS*
- [P1] **Caluã de Lacerda Pataca**, and Paula Dornhofer Paro Costa. (2019) Tipografia modulada pela fala. In: *Proceedings of the XII EADCA Encontro de Alunos e Docentes do DCA*.

COLLABORATIONS

Outside of collaborating with students and professors at my own university, I have established a global network of collaborators through networking at conferences and symposiums. Some of these collaborations have already borne fruit, and I have papers, published or forthcoming, that stem from work with researchers from Stanford University (US), University of Michigan (US), the University of Toronto (Canada), and the Gwangju Institute of Science and Technology (South Korea). Others are still in their initial stages, but I am hopeful they will be successful in the future. These include research groups from PXL University of Applied Sciences and Arts (Belgium), the University of Lübeck (Germany), and my alma mater, the University of Campinas (Brazil).

REFERENCES

Matt Huenerfauth, Dean and Professor at the Rochester Institute of Technology, USA

Ph.D. co-advisor · matt.huenerfauth@rit.edu

Roshan Peiris, Assistant Professor at the Rochester Institute of Technology, USA

Ph.D. co-advisor · Roshan.Peiris@rit.edu

Paula Dornhofer Paro Costa, Professor and Research Scientist at the University of Campinas, Brazil

M.Sc. advisor · paulad@unicamp.br

Luís Cláudio Portugal do Nascimento, Professor at the University of São Paulo, Brazil

BA professor · claudioportugal@usp.br